The secondary stage leading to demonstrable events such as precipitation, agglutination, lysis of cells, killing of live antigens, neutralization of toxins and other biologically active antigens, fixation of complement, immobilization of motile organisms and enhancement of phagocytosis.

The tertiary stage reactions lead to neutralization or destruction of antigens which might lead to tissue damage. They include humoral immunity against infectious disease as well as clinical allergy and other immunological disease.

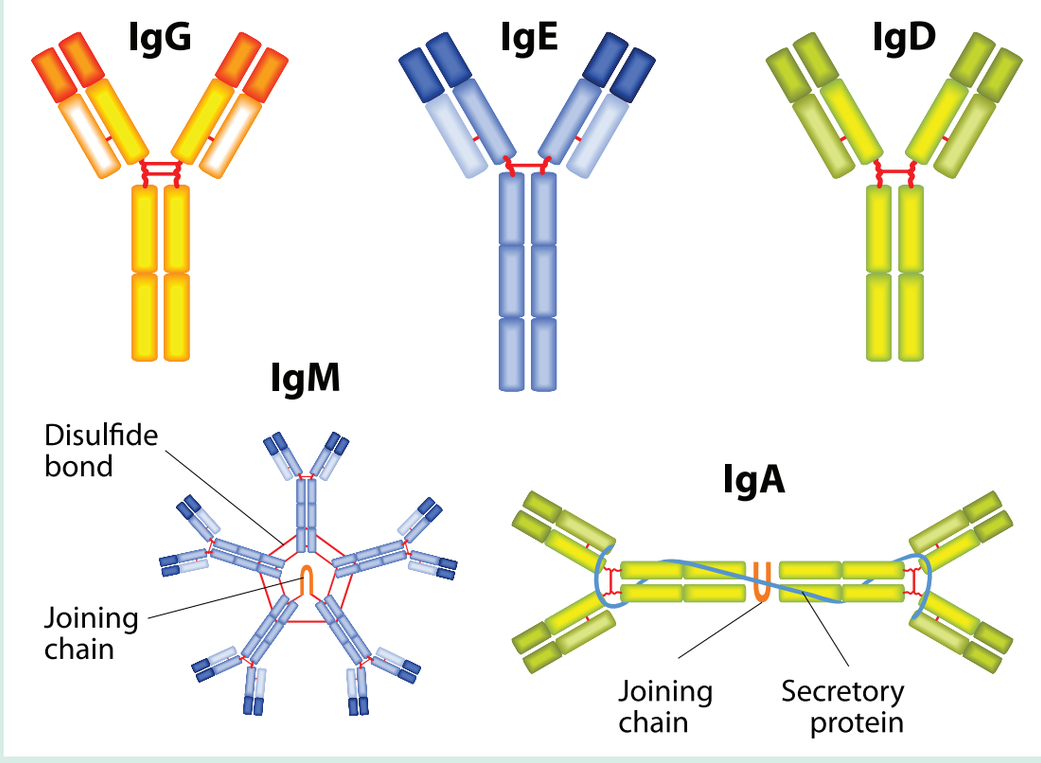
Antigen-antibody reactions have the following general characteristics:

1. The reaction is specific. However cross reaction may occur.
2. Entire molecules react and not fragment.
2. There is no denaturation of the antigen or the antibody during the reaction.
3. The combination occurs at the surface.
4. The combination is firm but reversible. The firmness of the union is influenced by the affinity and avidity of the reaction.

Affinity refers to the intensity of attraction between single epitope of the antigen and paratope of antibody molecules.

Avidity is the total strength of the bond after the formation of the antigen antibody complexes. Generally IgG possess greater affinity and IgM possess higher avidity

5. Both antigen and antibodies participate in the formation of agglutinates or precipitates.
6. Antigens and antibodies can combine in varying proportions.



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